

Frontend Implementation Guide

A simple guide to the screens, UI states, navigation, and interaction behavior shown in the PulseMind prototype.

How to use this document

Use the HTML prototype as the visual reference. Use this guide to understand what screens exist, what each interaction does, and how the important UI states change.

1. Purpose and implementation boundary

The prototype is an interactive design reference, not production frontend code. It contains three main screens and several UI states that are easier to understand by clicking through the prototype. This document makes those behaviors explicit so the frontend team does not need to reverse-engineer them.

- Visual styling, layout, spacing, typography, and component appearance should follow the HTML prototype.
- Navigation, selection behavior, review states, data-limited states, drawers, filters, and parameter switching should follow this guide.
- The production frontend should render values supplied by the real application/backend data flow.
- The frontend should not copy the prototype’s fake timers or scripted score changes as production logic.

Important: demo simulation vs. production behavior

The prototype automatically updates risk scores, timestamps, history charts, and some review states to demonstrate a live interface. In the real frontend, those changes should occur only when new application data/state is received.

2. Screen map

The prototype contains three primary screens.

Screen	Purpose	How it is reached
Patient Overview	Unit-level triage and patient selection.	Default / starting screen.
Patient Detail	Detailed review of one patient’s current respiratory-risk assessment.	Open the selected patient from Patient Overview.

Parameter Detail	Inspect one respiratory parameter and its charting provenance/history.	Select a parameter from Patient Detail.
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Navigation flow:

PATIENT OVERVIEW → PATIENT DETAIL → PARAMETER DETAIL

select patient open detail select parameter

3. Patient Overview

Purpose: provide a ranked unit-level view of adult ventilated ICU patients, highlight patients that need review, and let the user select one patient before opening the detailed assessment.

Main sections

- Header: PulseMind prototype/read-only status, live clock/date, user session information.
- Overview summary: pending-review count and connected-source count.
- Search + filters: Search Patient ID; All; Needs review; Data-limited.
- Triage board: ranked patient cards with risk level, score, data quality, review status, and primary driver.
- Selected-patient panel: 60-minute risk history, latest score, primary driver, and Open patient detail action.
- Input status: connected data-source/device status.
- Decision-support disclaimer.

Interactions and states

Element / state	Trigger	Expected behavior
Patient card selection	Click a patient card.	Select that patient and update the right-side/selected-patient panel. Do not navigate yet.
Open patient detail	Click "Open patient detail".	Navigate to Patient Detail for the currently selected patient.
Search	Type in Search Patient ID.	Filter the triage board by patient ID.
All filter	Select All.	Show all patients.
Needs review filter	Select Needs review.	Show only patients with Pending review status.
Data-limited filter	Select Data-limited.	Show only patients whose data is insufficient.
Empty result	Search/filter returns no patients.	Show an explicit empty-state message rather than a blank board.
Pending review	Patient warning status is Pending review.	Show pending-review styling/label and prioritize the patient in ranking.
Reviewed	The patient is not pending review.	Show Reviewed / Reviewed at time.
Insufficient data	Patient sufficient-data state is insufficient.	Display the warning that the risk score is not reliable.
Selected card	The patient is selected.	Apply selected-card emphasis and use that patient in the side panel.

Ranking rule shown in the prototype

The triage board is sorted first by review state (Pending review first), then by risk level, then by risk score.

4. Patient Detail

Purpose: give the clinician a detailed view of one patient's current PulseMind assessment, supporting factors, parameter values, explanation, and context.

Main sections

- Back to overview navigation.
- Selected-patient identity and assessment-refresh information.
- Warning/review-status panel.
- 60-minute respiratory-risk history.
- Current respiratory-risk score and risk band.
- Reading state: risk level, readings held at that level, imputed share, documentation share.
- Ranked factors contributing to the current score.
- Plain-language explanation.
- Guideline references.
- Respiratory parameter table.
- Patient context drawer.
- Decision-support disclaimer.

Review-status states

State	What is shown	Interaction
Pending review	A prominent warning panel stating that the reading has not been reviewed, plus patient/risk summary chips.	Show “Mark as reviewed”.
Reviewed	Reviewed status and, when available, the review time.	The mark-reviewed action is no longer required.
Mark as reviewed	The user activates the button.	Update the local UI review state to Reviewed without changing the risk score or ventilator settings.

Data-limited state

When the patient has insufficient data, the screen explicitly states that the risk score is not reliable. The prototype also withholds the plain-language explanation and guideline references for that reading.

Frontend rule

Do not leave the explanation/reference area looking like a normal successful state when data is insufficient. The unavailable state should be visually explicit.

Respiratory parameter table

Each row represents a respiratory parameter and is selectable.

Displayed information	Behavior	Displayed information
Parameter name + description	Identifies the parameter.	Parameter name + description

Latest value + unit	Shows the current point-in-time value.	Latest value + unit
Source	Shows measured, carried forward, or population reference provenance.	Source
Charting history / age	Indicates the recency/provenance of the value.	Charting history / age
Model use	Shows whether the parameter is a current score factor or simply available.	Model use
Row selection	Open Parameter Detail for the selected patient and parameter.	Row selection

Patient context drawer

The “View patient context” action opens a right-side drawer over the Patient Detail screen. It contains demographics, recorded medical history/comorbidities, and connected devices/sources. A close action returns to the underlying Patient Detail view.

- Drawer closed: off-screen; Patient Detail remains the active page.
- Drawer open: right-side panel visible with a background scrim.
- Close: hide the drawer and return to the same patient/detail state.
- If no comorbidities are recorded, show the prototype’s explicit “No recorded comorbidities” state.

5. Parameter Detail

Purpose: inspect one respiratory parameter in more detail, especially where its displayed value came from and how it has been charted over time.

Main sections

- Back to patient navigation.
- Parameter title and current patient/chart time.
- Parameter navigation chips for switching between parameters.
- Current value, unit, current source, last measured time, and model-use state.
- Charting history/provenance chart.
- Provenance summary.
- Conditional population-reference warning.
- Parameter metadata.
- Decision-support disclaimer.

Parameter switching

The parameter chips at the top of the page switch the content in-place to the selected parameter while keeping the same patient. The active parameter is visually highlighted.

Source states

Source state	Meaning shown by the prototype	UI behavior
Measured	Read from the device at the reading's chart time.	Display measured provenance.
Carried forward	The last measured value is being reused rather than re-measured.	Display carried-forward provenance and retain the last measured time.
Population reference	A population value, not measured on this patient.	Use distinct provenance styling and show the population-reference explanatory panel.

Chart interaction

The chart displays parameter charting provenance over the recent time window. Hovering a chart point shows a tooltip containing the value/time/source and additional provenance detail where relevant. The chart is about charting provenance, not a PulseMind-generated clinical trend classification.

6. Global state reference

The following states recur across the prototype and should be implemented consistently.

State	Where it appears	Frontend expectation
Risk level	Overview + Patient Detail	Use a consistent label/color treatment for Low, Medium, High, and Critical.
Pending review	Overview + Patient Detail	Visually distinguish from reviewed readings and surface the review action on Patient Detail.
Reviewed	Overview + Patient Detail	Show Reviewed and the time when available.
Insufficient data	Overview + Patient Detail	Clearly state that the risk score is not reliable; suppress explanation/references on Patient Detail.
Measured provenance	Patient Detail + Parameter Detail	Identify the value as measured.
Carried-forward provenance	Patient Detail + Parameter Detail	Identify reused values and preserve last-measured context.

Population-reference provenance	Patient Detail + Parameter Detail	Clearly state that the value was not measured on the patient.
Selected item	Overview + Parameter Detail	Visually highlight the active patient/parameter.
No result	Overview	Show an explanatory empty state when search/filter returns nothing.
Drawer open/closed	Patient Detail	Overlay context drawer without changing the selected patient.

7. Navigation and interaction quick reference

From	User action	Result
Patient Overview	Click patient card	Select patients only.
Patient Overview	Click Open patient detail	Open Patient Detail for selected patients.
Patient Detail	Click Back to overview	Return to Patient Overview.
Patient Detail	Click View patient context	Open patient-context drawer.
Patient Detail	Click Close / dismiss drawer	Close drawer; remain on Patient Detail.
Patient Detail	Click Mark as reviewed	Change review state to Reviewed.
Patient Detail	Click parameter row	Open Parameter Detail for that patient + parameter.
Parameter Detail	Click parameter chip	Switch displayed parameter; keep same patient.
Parameter Detail	Hover chart point	Show chart tooltip.
Parameter Detail	Click Back to patient	Return to Patient Detail for same patient.

8. What this prototype does not define

The prototype is a design/handoff reference. The following production concerns are not fully defined by the HTML and should therefore not be invented by the frontend team without a separate decision or data contract.

- Backend/API endpoints and request/response formats.

- Authentication implementation beyond the displayed session/sign-out UI.
- Persistence rules for review actions.
- Production refresh/polling/WebSocket strategy.
- Loading, backend-error, network-offline, permission-denied, and retry states.
- Final clinical thresholds or model logic beyond the values currently displayed in the prototype.
- Production device-integration logic.
- Accessibility requirements and responsive breakpoints beyond the behavior visible in the prototype.

Recommended handling of undefined states

Where a required production state is missing from the prototype, mark it for design/product clarification rather than silently choosing a behavior.

9. Frontend handoff checklist

- ☐ Implement the three primary screens and preserve patient/parameter context during navigation.
- ☐ Implement Overview search, All / Needs review / Data-limited filters, selection, ranking, and empty results.
- ☐ Implement Pending review, Reviewed, and Insufficient data states.
- ☐ Implement Patient Detail score/history, reading state, ranked factors, explanation/references, and respiratory-parameter table.
- ☐ Implement the patient-context drawer and its open/close behavior.
- ☐ Implement Parameter Detail switching, provenance states, population-reference warning, and chart hover tooltip.
- ☐ Keep measured / carried-forward / population-reference provenance visibly distinct.
- ☐ Do not reproduce prototype-only fake timers or scripted risk escalation in production.
- ☐ Bind dynamic content to the agreed frontend/backend data contract.
- ☐ Raise missing production states (loading/error/offline/etc.) for explicit product/design decisions.